Biodiversity Genomics Europe

WP12 Joint Network Activities

T12.2-3-7 - Pollinator Intraspecific Genetic Diversity Substream

Standard Operating Procedure

Contributing authors: João Pimenta, Sónia Ferreira, José Melo-Ferreira and Robert Waterhouse

Funded by the EU's Horizon Europe Research and Innovation Action
https://biodiversitygenomics.eu/

Table of Contents

Funded by the EU's Horizon Europe Research and Innovation Action

Introduction and Objectives

The Biodiversity Genomics Europe **(BGE) Project** has the overriding aim of accelerating the use of genomic science to enhance understanding of biodiversity, monitor biodiversity change, and guide interventions to address its decline. The **BGE Project** comprises activities focused on DNA Barcoding (**Barcoding Stream**) and Reference Genome Generation (**Genomes Stream**) for eukaryotic species across Europe, bringing together two European networks: BIOSCAN Europe and the European Reference Genome Atlas **(ERGA)**. The **Genomes Stream** aims to establish and implement large-scale biodiversity genomic data generation pipelines to accelerate the production and accessibility of reference-quality, complete, genome sequences for species across the whole of European biodiversity that will support applications in the fields of biodiversity characterisation, conservation, and biomonitoring. The **Genomes Stream** focuses on generating reference-quality genomes from critical European biodiversity, biodiversity hotspots, pollinators, and a selection of applied case studies.

The present **Standard Operating Procedure (SOP)** supports the development of the **Pollinator Intraspecific Genetic Diversity (PID)** substream integrated in **Work Package 12 (WP12)**, which connects the **Barcoding** and **Genomes** Streams of the BGE Project. The aim of the **PID substream** is to design monitoring tools that capture intraspecific genetic compositions, an often neglected Essential Biological Variable in biomonitoring. Focused on three pollinator species, it will combine analyses of full reference genomes and population-level whole genome resequencing data to develop and validate multi-locus assays capable of capturing genetic diversity across species. It is composed of a series of sequential tasks that will ensure Pollinator Population Sampling (Task 12.2), Laboratory work and Data collection (Task 12.3) and Data analyses and Application Development (Task 12.7). Collectively, the PID substream is led by *Centro de Investigação em Biodiversidade e Recursos Genéticos* (CIBIO - Portugal) with partners from *Consejo Superior de Investigaciones Científicas* (CSIC - Spain), *University of Primorska* (UP - Slovenia), *Aristotle University of Thessaloniki* (AUTH - Greece), and *University of Oslo* (UiO - Norway). This SOP provides a guide for species identification, capturing and collection of metadata, following the regulations of the BGE project (**Sampling Code of Conduct** and **Sampling Metadata Manifest**).

 Funded by the EU's Horizon Europe Research and Innovation Action

Model Species and Sampling Design

The PID substream will focus on three model species, widely distributed across Europe, representing three important pollinator groups:

Butterfly: *Maniola jurtina* (Meadow brown)

Hoverfly: *Syritta pipiens* (Thick-legged hoverfly)

Bumblebee: *Bombus terrestris* (Buff-tailed bumblebee)

The sample scheme was designed to capture the genetic diversity of the three pollinator species across Europe. The target is to capture between 25-30 specimens, per **geographic locality**, collected in individual plastic tubes and immersed in ethanol 96-100%. The definition of population/locality will vary depending biological characteristics of the species captured:

- Butterfly: *Maniola jurtina* (Meadow brown)

  A population can be considered the group of butterflies living in a flower/grass/forest patch of <1km^2.

- Hoverfly: *Syritta pipiens* (Thick-legged hoverfly)

  As for the butterflies, for hoverflies a population can be considered the group of hoverflies living in a flower/grass/forest patch of <1km^2.

- Bumblebee: *Bombus terrestris* (Buff-tailed bumblebee)

  Whenever possible, samples for *B. terrestris* should be collected at least 300m apart to minimise the probability of collecting specimens from the same colony, due to the clonal nature of the species (Wolf & Moritz, 2008, 10.1051/apido:2008020). Please note that we are focusing the sampling efforts on diploid individuals (females - workers), in order to capture the genetic diversity of the populations.

Natural environments

Natural sites should be considered as locations at least 10 km away from urban sites and land-use maps should be dominated by agricultural and semi-natural/natural cover, with low density of roads and residential cover. A GIS software should be used to identify natural areas.

Urban environments

Urban locations can be considered gardens and parks within cities, surrounded by a high density of human infrastructure (buildings, roads, etc).

 Funded by the EU's Horizon Europe Research and Innovation Action

https://biodiversitygenomics.eu/

Maniola jurtina (*Lepidoptera*, Meadow brown)

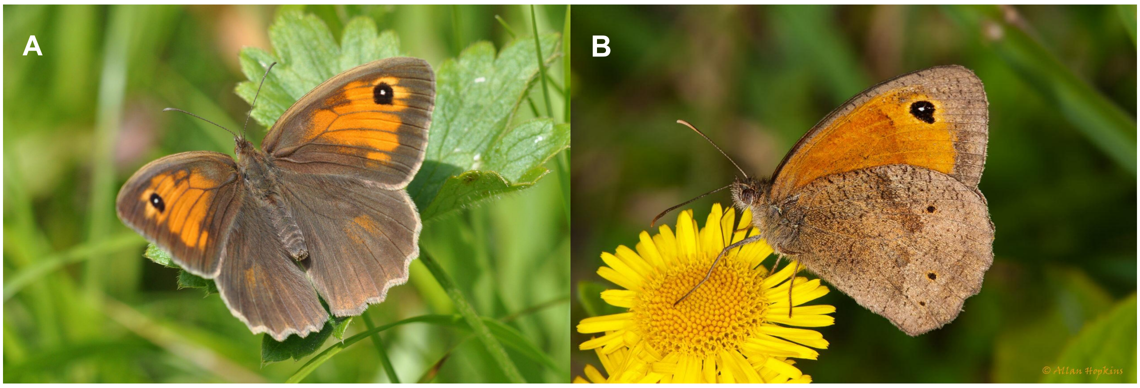

Figure 1. A) Dorsal view of *M. jurtina*; credits: Markus Dumke, some rights reserved (CC BY-NC-ND). B) Lateral view; credits: Allan Hopkins, some rights reserved (CC BY-NC-ND).

Reference genome:

Complete: 402Gb

Genome note: https://wellcomeopenresearch.org/articles/6-296/v1

Ensembl: https://rapid.ensembl.org/Maniola_jurtina_GCA_905333055.1/Info/Index

Classification

Class: *Insecta*

Order: *Lepidoptera*

Family: *Nymphalidae*

Genus: *Maniola*

Species: *M. jurtina*

Morphology

- *Maniola jurtina* can reach a wingspan of 50 mm with females being larger.
- The body, eyes and antennae are brown (or smoky brown). Females are usually lighter coloured than males.
- The upper surface of the wings is brown (or smoky brown) with a lighter submarginal band.
- The forewings have a subapical black ocellus, with a white pupil in the centre. Males diverge from females, as they have an extensive, well-marked androconial spot. The underside of the forewings is orange, with a greyish-brown marginal edge and a black ocellus with a white pupil; the underside of the hind wings is brown.

 Funded by the EU's Horizon Europe Research and Innovation Action

https://biodiversitygenomics.eu/

- In the upper surface of the forewings, females have more larger orange spots, with the subapical ocellus being visibly larger than in males.

Distribution and Habitat

- Widespread, common and abundant in Europe. Spread from North Africa to southern Scandinavia (but expanding northwards). It inhabits Mediterranean islands (Canary Islands) and the British Isles (**Fig. 2**).

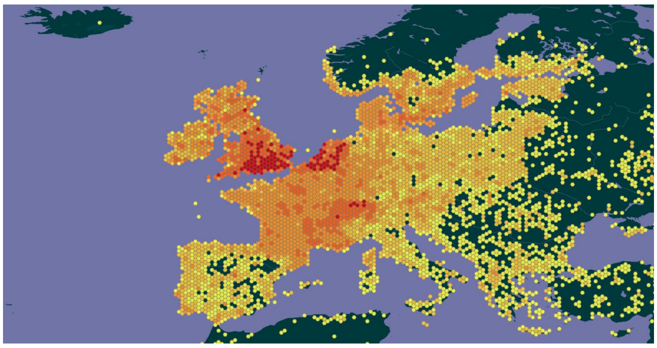

Figure 2. *M. jurtina* distribution based on data occurrence from GBIF.

- It can be found in fry or damp conditions, and up to altitudes of 1500m. The species colonises extensive meadows and pastures, and it is common for example in limestone grasslands, on flood embankments or in humid meadows around wetlands. While the species can survive in extensively managed grasslands, it is less common or even absent in intensive agricultural land.

Funded by the EU's Horizon Europe Research and Innovation Action
https://biodiversitygenomics.eu/

Syritta pipiens (*Diptera*, Thick-legged hoverfly)

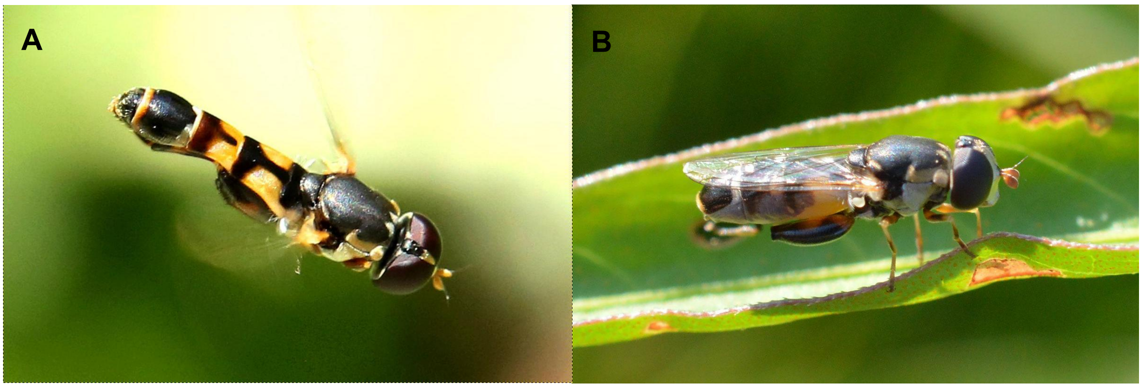

Figure 3. A) Dorsal view of *S. pipiens;* credits: Peter Chen, some rights reserved (CC BY). B) Lateral view; credits: Dimitar Boevski, some rights reserved (CC BY).

Reference genome

Assembly submitted - 315 Mb

ENA: https://www.ebi.ac.uk/ena/browser/view/GCA_905147025

Taxonomy

Class: *Insecta*

Order: *Diptera*

Family: *Syrphidae*

Genus: *Syritta*

Species: *S. pipiens*

Morphology

- *Syritta pipiens* is a small syrphid, 6 to 9 mm long.
- It has a white face and brownish or pinkish antennae with black edges.
- The thorax is black and the abdomen is slender, black with yellow spots on tergites T2, T3 and T4. The posterior femur is enlarged with a row of spines along the ventral edge, with a light spot in the middle.
- The wings have a transversal rib.
- It presents sexual dimorphism: in males, the eyes are contiguous (holotypes); in females, the eyes are wide apart. Males are strongly territorial.

Distribution and habitat

 Funded by the EU's Horizon Europe Research and Innovation Action

- Widespread common and abundant in Europe (**Fig. 4**)
- Anthropophilic species that can be found in urban environments (gardens, parks, etc) and farms. It can also be found in wetlands, edges of bogs and near freshwater bodies.

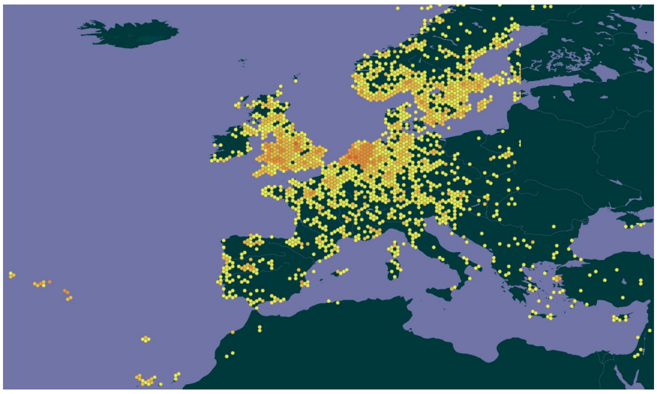

Figure 4. *S.pipiens* distribution based on data occurrence from GBIF.

Funded by the EU's Horizon Europe Research and Innovation Action

Bombus terrestris **(*Hymenoptera*, Buff-tailed bumblebee)**

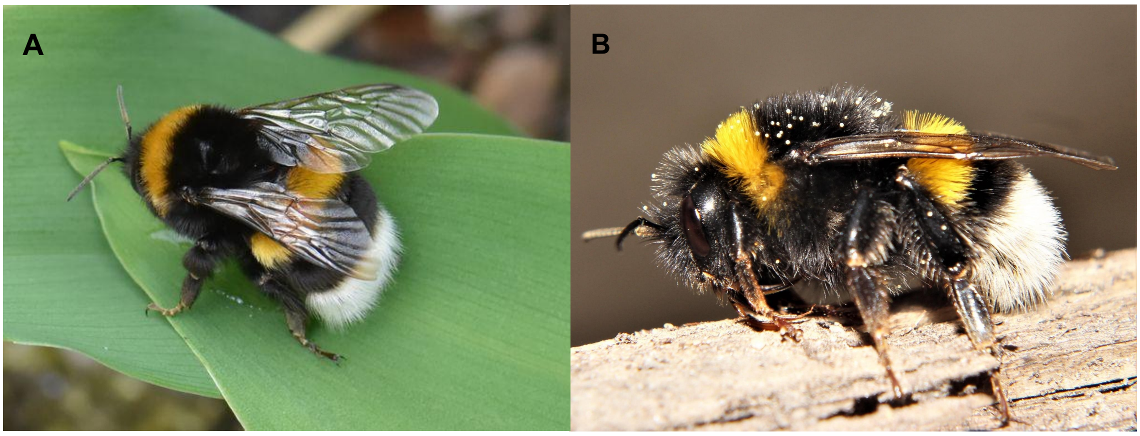

Figure 5. A) Dorsal view of *B. terrestris*; credits: Herman Berteler, some rights reserved (CC BY-NC). B) Lateral view; credits: Bill Lucas, some rights reserved (CC BY-NC).

Reference genome

Annotation complete - 248 Mb

ENSEMBL: https://rapid.ensembl.org/Bombus_terrestris_GCA_000214255.1/Info/Index

Taxonomy

Class: *Insecta*

Order: *Hymenoptera*

Family: *Apidae*

Genus: *Bombus*

Species: *B. terrestris*

Morphology

- The adult can reach from 11mm (workers) to 22mm (queen) long.
- The body is covered with hair, arranged in parallel lines of black, yellow and white stripes. The species has 9 subspecies described (8 present in Europe), mainly distinguished by the stripe pattern of the body (**Fig. 6**)
- It also has two yellow bands, one in the chest and one in the abdomen. It lives in communities of 20 to 150 individuals of different types (queen, males and workers).
- Workers may be similar to ***Bombus lucorum***.

Distribution and habitat

- Widespread common and abundant in Europe (**Fig. 7**).

- The species has a broad geographic and climatic distribution and it inhabits a wide range of habitats including cosmopolitan environments.

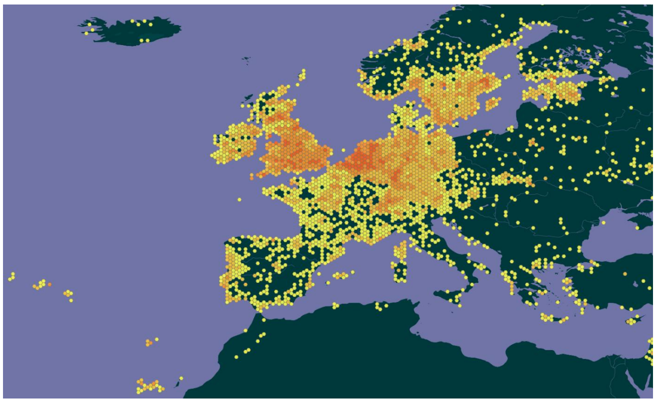

Figure 6. *B. terrestris* distribution based on data occurrence from GBIF.

Sampling procedures

Pollinator insects are a diverse group of invertebrates spreading through several taxonomic groups, with very different rates of abundance, life cycles and phenologies. Several techniques can be used to sample these groups, and more than one technique can be used to capture the same species. As all sampling methods have their own limitations, usually at least two sampling strategies are used to capture a representative sample of a species. The choice of sampling procedure varies depending on the biology, preferred habitat, and activity patterns of the species. In the PID Substream the sampling focus is on three insect flying species: a butterfly - *Maniola jurtina* (Meadow Brown), a hoverfly - *Syritta pipiens* (Thick-legged hoverfly) and a Bumble bee - *Bombus terrestris* (Buff-tailed bumble bee). To sample these a few sampling techniques can be used, namely **direct search, aerial nets, and sweep nets. The use of Pan-traps is also very common for pollinator sampling, however it implies a lot of other species to be collected as by-catch and the efficiency towards the target species is low.** Suitable sampling techniques are briefly described (for more details check Ferreira S. et al (2018))

Direct search

Usually considered a low efficiency technique due to the low number of specimens captured in total, it consists in the direct observations of surfaces in environments where we expect to find the targeted species. It largely depends on previous knowledge of the species life history and habitat preferences. <u>All three species in this project can be collected with this method. Due to their local abundance and conspicuity this method provides good results and allied with aerial and sweep nets is likely to be the most efficient strategy to obtain the samples for the present study</u>.

Aerial nets

Also known as butterfly nets, aerial nets can be used to collect flying insects. Usually, aerial nets are composed of a bag made of meshed and lightweight material fixed to a wire hoop, and fixed to a wooden or metal pole. This method implies the active search and chase of specimens. <u>All three species in this project can be collected with this method</u>.

Sweep nets

This method is used to sweep insects out of riparian vegetation. Depending on the shape and size of the net, results can vary. In this method, the collector carries out random sampling in the vegetation (ideally short vegetation) and specimens are selected and collected afterwards. Sweep nets are made of cotton Muslin or sailcloth. This technique is particularly useful to sample *Syritta pipiens* (Thick-legged hoverfly), as due to its smaller size and tendency to be perching on herbaceous vegetation it is less likely to be detected by direct observation.

Funded by the EU's Horizon Europe Research and Innovation Action
https://biodiversitygenomics.eu/

Specimen Preservation and Shipping

Insect preservation method

- After capturing the specimen, the insect should be **preserved individually in labelled tubes** immersed in **ethanol 96-100%**.
- All metadata should be acquired accordingly with the Sampling Metadata Manifest of ERGA (see details below).
- The label in tubes is a unique standardised identifier of the individual and must be kept in the metadata sheet. The unique ID nomenclature should include the acronym for **country of collection** (e.g. Portugal - **PT**) followed by _underscore and the acronym for **the species** (e.g *Bombus terrestris* - **BT**). The acronyms are followed by _underscore RUNNING NUMBERS. Specimen_IDs must be unique to an individual (e.g., PT_BT_5 cannot be used again after it has been assigned to a specimen).
- **Hoverflies** should be preserved in **2ml** tubes while the **butterflies** and **bumblebees** should be preserved in **8ml** tubes with good sealing properties.
- Specimens can then be preserved at room temperature or stored in the freezer (-20ºC).

Shipping

Samples should be shipped using an express courier, only after the Material Transfer Agreement (MTA) has been signed by both the supplier and recipient institutions (see below for details). The tubes should be placed in boxes with individual support for each tube inside a larger box of styrofoam. Outside the box should be attached the label UN3373, for diagnostic specimen.

Samples should be address to:

A/C: João Pimenta (CIBIO)
Campus Agrário de Vairão
Rua Padre Armando Quintas, nº 7
4485-661 Vairão
Portugal
Tel: +351 252660420

Funded by the EU's Horizon Europe Research and Innovation Action
https://biodiversitygenomics.eu/

Regulations and procedures

Sampling code of conduct and metadata manifest

By being a supplier of specimens that will be integrated in the **BGE Project Genomes Stream**, the sample provider as well as all members of the **Case Study Team** agree to abide by the **Sampling Code of Practice**, and are responsible for ensuring legal compliance of sampling. The definition of **Case Study Team** is detailed below. The sample provider is also responsible for the collection of all required **Metadata** for each specimen.

Material transfer agreement

To formally establish the research collaboration for the use of the biological material, an agreement will be established between the supplier institution and the receiver institution. The **Material Transfer Agreement** covers all the conditions that both institutions will agree upon for the temporary transfer of biological material containing genetic resources to be used the the **Pollinator Intraspecific Genetic Diversity Substream** with no change in ownership/permanent custodianship.

Case Study Team

The **Case Study Team Definition** for the Genomes Stream of the BGE Project pertains to applications (case studies) developed as part of the BGE Project Genomes Stream. It is an extension of the **ERGA Genome Team Definition**. The case study team definition describes a set of roles, delivered by individuals or collectives, and it is agreed that delivery of these roles is sufficient engagement to warrant authorship on the emerging publications. Once samples are shipped to CIBIO the list of members of the collecting team should be sent to the coordination of the project.

Funded by the EU's Horizon Europe Research and Innovation Action